

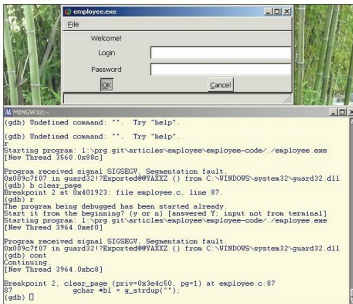


Figure 1: Debugging with GDB

Automatic installation of MinGW

Go to <http://www.mingw.org>. Download and run the automated MinGW Installer (as of writing, this is MinGW-5.1.6.exe). From the package list, install at least the *MinGW Base tools* and *make* packages. Other components are optional for our application. The installer connects to the server, and installs the chosen components.

Manual installation of MinGW

Our other option is to download and install components individually. If you want to do that, you can find the required files on the accompanying CD. Otherwise, you can find documentation on how to do this at http://www.mingw.org/wiki/Getting_Started.

Installing MSYS

Now that you have all the tools set up, what you need next is a shell. As described earlier, install the *MSYS* and *msysDTK* packages. When asked about where exactly you've installed *minGW*, give your *minGW* location. Once installed, please go through the *readme* file. Start the shell from the Start menu.

Working with MSYS

In MS Windows, clicking *Start button* -> *Programs* -> *minGW->MSYS* will give you *MSYS* and *MSYS(rxt)*. It is important that you understand the basics of working with *MSYS*, especially if you have worked on *Cygwin*, because these two environments are inherently different. One cannot use *Cygwin* executables in *MSYS*, and vice-versa. Here are a few tips to make your life with *MSYS* easy:

- If you want to execute a program (say, *OpenOffice.org*) from the command line, you will need to create an alias or a script that invokes the program. The alias would be somewhat as follows:

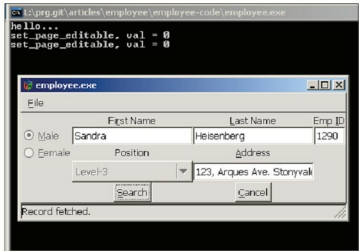


Figure 2: Background 'debug console' window

```
soffice="/c:/Program Files/OpenOffice.org/2.3/program/soffice.exe"
$@
```

- Or, you could create a script like what's shown below:

```
$ cat > soffice
start "/c:/Program Files/OpenOffice.org/2.3/program/soffice.exe"
$@
```

- For any native executable to work, the installed path should be in the *PATH* variable. (Or else, install it in *c:\mingw\{bin,usr\bin,usr\local\bin}* etc.)
- Simply selecting the data with a mouse click/drag will copy the data to the clipboard (in *MSYS rxvt*). To get the current clipboard contents, do a *\$cat /dev/clipboard* (you cannot *\$ls /dev*, though).

Installing Glade/GTK+

As mentioned earlier, in the first part of this tutorial series, Glade has a Windows port. Let's install Glade-3 and GTK+ on Windows. For GTK+, go to <http://www.gtk.org/download-windows.html> and download http://ftp.gnome.org/pub/gnome/binaries/win32/gtk+/2.20/gtk+-bundle_2.20.0-20100406_win32.zip. Unzip this into the *c:\minGW* directory. Download Glade-3 from <http://ftp.gnome.org/pub/GNOME/binaries/win32/glade3/3.6/glade3-3.6.7-with-GTK+.exe>. Although GTK+ is already installed, I could not find a Glade Windows installable without GTK+. The other option is to recompile Glade, which is beyond the scope of this article.

Checking for tools

Copy the source files we have already created in Linux, in the previous exercises we went through in this series, to your home directory under *employee-code*. If you connect a removable drive (say, under *L:*), you will have to mount it explicitly if */L:* is not available. For example:

```
om@Desktop:~$} mount
```